

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	<i>Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)</i>		
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Section / Batch:	AI & DS	Date:	23-07-26

Instructions to Students

- Answer ALL questions. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1 Q2,Q3	40 60 (2X30)
GRAND TOTAL:			100 Marks

SECTION A – APPLICATION BASED QUESTIONS

Q1. Closest Pair – Simple Coordinate Set

Given the set of points $\{(1,2), (4,6), (5,5), (9,1), (2,3)\}$, apply the Closest Pair Brute Force algorithm. Clearly interpret the problem and justify the method selection. Compute the distance between every possible pair and show the calculations clearly. Identify the closest pair and count the total number of comparisons. Analyze the time complexity of the brute force approach. Verify the correctness of the result and interpret its usefulness in geometric applications.

Q2. Binary Search – Search for Central-Upper Element

Given the sorted dataset $[5, 10, 15, 20, 25, 30, 35, 40, 45]$, apply Binary Search to find the element 35. Clearly interpret the problem and justify why Binary Search is suitable for this ordered dataset. Show the sequence of middle elements examined and the reduction of the search interval. Count the total number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret the performance of the algorithm.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (40 Marks)	4	4	4	4	4	40
Q2 (30 Marks)	4	4	4	4	4	30
Q3 (30 Marks)	4	3	3	2	2	22
Overall Total (100)						92

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Closest Pair – Simple Coordinate Set

Given the set of points $\{(1,2), (4,6), (5,5), (9,1), (2,3)\}$, apply the Closest Pair Brute Force algorithm. Clearly interpret the problem and justify the method selection. Compute the distance between every possible pair and show the calculations clearly. Identify the closest pair and count the total number of comparisons. Analyze the time complexity of the brute force approach. Verify the correctness of the result and interpret its usefulness in geometric applications.

Problem Interpretation

We are given the set of points:

$\{(1,2), (4,6), (5,5), (9,1), (2,3)\}$

The objective is to find the two points having the **minimum Euclidean distance** using the **Brute Force Closest Pair Algorithm**.

Why Brute Force is Suitable

The brute force approach is suitable because:

- The dataset contains only **5 points**, making computation simple.
- It compares **every possible pair**, ensuring the exact closest pair is found.
- No sorting or preprocessing is required.

Step 1: Formula Used

The Euclidean distance between two points

$P1(x1,y1), P2(x2,y2)$

Is

$$d = \sqrt{(x2-x1)^2 + (y2-y1)^2}$$

Step 2: List the Points

Point	Coordinates
P ₁	(1,2)
P ₂	(4,6)
P ₃	(5,5)
P ₄	(9,1)
P ₅	(2,3)

Number of points:

$$n=5$$

Total possible pairs:

$$n(n-1) / 2 = 5(4) / 2 = 10$$

Therefore, 10 comparisons are required.

Step 3: Calculate Distance Between Every Pair

Pair 1: P₁ and P₂

$$\begin{aligned}
 &= \sqrt{(4-1)^2 + (6-2)^2} \\
 &= \sqrt{3^2 + 4^2} \\
 &= \sqrt{9 + 16} \\
 &= \sqrt{25} \\
 &= 5
 \end{aligned}$$

Pair 2: P₁ and P₃

$$\begin{aligned}
 &= \sqrt{(5-1)^2 + (5-2)^2} \\
 &= \sqrt{4^2 + 3^2} \\
 &= \sqrt{16 + 9} \\
 &= \sqrt{25} \\
 &= 5
 \end{aligned}$$

Pair 3: P₁ and P₄

$$\begin{aligned}
 &= \sqrt{(9-1)^2 + (1-2)^2} \\
 &= \sqrt{8^2 + (-1)^2} \\
 &= \sqrt{64 + 1}
 \end{aligned}$$

$$= \sqrt{65}$$

$$\approx 8.06$$

Pair 4: P₁ and P₅

$$= \sqrt{(2-1)^2 + (3-2)^2}$$

$$= \sqrt{1+1}$$

$$= \sqrt{2}$$

$$\approx 1.414$$

Pair 5: P₂ and P₃

$$= \sqrt{(5-4)^2 + (5-6)^2}$$

$$= \sqrt{1+1}$$

$$= \sqrt{2}$$

$$\approx 1.414$$

Pair 6: P₂ and P₄

$$= \sqrt{(9-4)^2 + (1-6)^2}$$

$$= \sqrt{25+25}$$

$$= \sqrt{50}$$

$$\approx 7.071$$

Pair 7: P₂ and P₅

$$= \sqrt{(2-4)^2 + (3-6)^2}$$

$$= \sqrt{4+9}$$

$$= \sqrt{13}$$

$$\approx 3.606$$

Pair 8: P₃ and P₄

$$= \sqrt{(9-5)^2 + (1-5)^2}$$

$$= \sqrt{16+16}$$

$$= \sqrt{32}$$

$$\approx 5.657$$

Pair 9: P₃ and P₅

$$= \sqrt{(2-5)^2 + (3-5)^2}$$

$$= \sqrt{9+4}$$

$$= \sqrt{13}$$

$$\approx 3.606$$

Pair 10: P₄ and P₅

$$\begin{aligned}
 &= \sqrt{(2-9)^2 + (3-1)^2} \\
 &= \sqrt{49+4} \\
 &= \sqrt{53} \\
 &\approx 7.280
 \end{aligned}$$

Step 4: Summary Table

Pair	Distance
(1,2) – (4,6)	5.000
(1,2) – (5,5)	5.000
(1,2) – (9,1)	8.062
(1,2) – (2,3)	1.414
(4,6) – (5,5)	1.414
(4,6) – (9,1)	7.071
(4,6) – (2,3)	3.606
(5,5) – (9,1)	5.657
(5,5) – (2,3)	3.606
(9,1) – (2,3)	7.280

Step 5: Identify the Closest Pair

Minimum distance obtained:

$\sqrt{2} \approx 1.414$

There are two closest pairs:

- (1,2) and (2,3)
- (4,6) and (5,5)

Both have the same minimum distance.

Step 6: Number of Comparisons

For $n=5$:

$$n(n-1) / 2 = 10$$

Total comparisons = 10

Step 7: Time Complexity Analysis

The brute force algorithm checks every pair.

Number of comparisons:

$$n(n-1) / 2$$

Ignoring constants,

$$O(n^2)$$

Complexity

- Best Case: $O(n^2)$
- Average Case: $O(n^2)$
- Worst Case: $O(n^2)$

The algorithm always examines every pair.

Step 8: Correctness Verification

All 10 possible pairs were examined exactly once.

The smallest distance found is

$$\sqrt{2} \approx 1.414$$

Hence, the algorithm correctly identifies the closest pair(s).

Step 9: Interpretation and Applications

The Closest Pair algorithm is useful in many real-world applications such as:

- Geographic Information Systems (GIS)
- GPS navigation
- Computer graphics

- Robotics
- Air traffic control
- Pattern recognition
- Machine learning (clustering)

The brute force method is simple and guarantees the correct answer but becomes inefficient for very large datasets because of its quadratic time complexity.

Final Answer

- **Closest distance:** $\sqrt{2} \approx 1.414$
- **Closest pairs:** (1,2) and (2,3) , d (4,6) and (5,5)
- **Total comparisons:** 10
- **Time complexity:** $O(n^2)$

Binary Search – Search for Central-Upper Element

Given the sorted dataset [5, 10, 15, 20, 25, 30, 35, 40, 45], apply Binary Search to find the element 35. Clearly interpret the problem and justify why Binary Search is suitable for this ordered dataset. Show the sequence of middle elements examined and the reduction of the search interval. Count the total number of comparisons made. Analyze the time complexity of the method. Verify the correctness of the result and interpret the performance of the algorithm.

Problem Interpretation

Given the sorted array

[5, 10, 15, 20, 25, 30, 35, 40, 45]

Find the element

35

using the **Binary Search Algorithm**.

Why Binary Search is Suitable

Binary Search is appropriate because:

- The array is already **sorted in ascending order**.
- Binary Search repeatedly divides the search interval into two halves, making it much faster than linear search.
- It efficiently locates the target with logarithmic time complexity.

Step 1: Initial Array

Index	0	1	2	3	4	5	6	7	8
Value	5	10	15	20	25	30	35	40	45

Target = **35**

Initial values:

- Low = 0
- High = 8

Step 2: First Iteration

$$\text{Mid} = 0 + 8 / 2 = 4$$

Value at index 4:

25

Compare:

$$35 > 25$$

Search the **right half**.

New interval:

- Low = 5
- High = 8

Step 3: Second Iteration

$$\text{Mid} = 5 + 8 / 2 = 6$$

Value at index 6:

35

Comparison:

35=35

Element found.

Step 4: Sequence of Middle Elements Examined

The middle elements checked are:

1. **25**
2. **35**

Sequence:

25→35

Step 5: Reduction of Search Interval

Iteration	Low	High	Mid	Middle Element	Result
1	0	8	4	25	Search right half
2	5	8	6	35	Element found

Step 6: Total Comparisons

Comparison 1:

35 vs 2535

Comparison 2:

35 vs 3535

Therefore,

$$\text{Total Comparisons} = 2$$

Step 7: Time Complexity Analysis

At every step, Binary Search halves the search space.

Time complexity:

- **Best Case:** $O(1)$ (target is the first middle element)
- **Average Case:** $O(\log n)$
- **Worst Case:** $O(\log n)$

For this dataset:

$$n=9$$

Maximum iterations:

$$\lceil \log_2(9) \rceil = 4$$

Only **2 iterations** were required.

Step 8: Correctness Verification

The element **35** is located at:

Index 6

The algorithm correctly identifies the target after two comparisons.

Step 9: Interpretation of Performance

Binary Search is highly efficient because:

- It reduces the search interval by half after each comparison.
- For large datasets, it performs significantly fewer comparisons than Linear Search.
- It is widely used in databases, search engines, operating systems, and sorted data structures.

Final Answer

- **Target element found:** 35
- **Index:** 6
- **Sequence of middle elements examined:** 25→35
- **Total comparisons:** 2
- **Time complexity:** Best Case : $O(1)$, Average/Worst Case : $O(\log n)$

The Binary Search algorithm successfully locates the target element by repeatedly halving the search interval, demonstrating its efficiency for searching in sorted datasets.